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CSA15 Cloud Computing and Big Data Analytics

CO1 AT2 - Digital Concept Map and Service Model Matrix

Individual Student Submission Template

Student and Faculty Details

Student Name	L Nihas Goud	Register Number	192472346
Class / Section	CSE(AI)	Slot	B
Course Code	CSA1512	Course Title	Cloud Computing and Big Data Analytics
Capstone Title	SkillHire Cloud Resume-to-Job Matcher Using NLP Skill Gap Intelligence		
Submission Date	13-07-2026		
Faculty Name	B.RAJESH BABU	Employee ID	
Department	CSE(AI)		

Assessment Brief

Course Outcome	CO1 - Explain the evolution, architecture, characteristics, service models, and emerging paradigms of cloud computing.
Assessment Type	Individual concept mapping, service-model responsibility analysis, and capstone-cloud interpretation.
Assessment Focus	Each map must show key nodes, link labels, relationship meaning, and a logical hierarchy. The response must connect cloud fundamentals to the student's own perception of the capstone title.
Submission Mode	Editable DOCX + exported diagram evidence + GitHub repository link submitted in Google Classroom.
Permitted Tools	draw.io / diagrams.net, Lucidchart, Miro, Canva, PowerPoint, Google Drawings, or equivalent diagramming tool.
Individual Rule	Every student must submit their own concept map, matrix, capstone-cloud mapping, and reflection. The capstone title may be shared by a team, but the assessment answer must not be common.
Total Marks	100 marks

Code of Conduct and Student Assurance

Academic Integrity Statement

This assessment must be completed individually. Students may discuss the capstone domain with their team, but the concept map, service model matrix, explanations, GitHub evidence, and reflection submitted here must be individually prepared.

Student Assurance	I confirm that this submission represents my own understanding of CO1 concepts and my individual interpretation of the selected capstone title.
Tool Use	I have used digital tools only for diagram preparation, organization, and presentation. I have not copied another student's diagram, matrix, or explanation.
Evidence	I have uploaded the required files to my GitHub repository and submitted the repository link / evidence link through Google Classroom.
Student Signature	L Nihas Goud
Date	13-07-2026

Procedure for Preparing the Digital Concept Map

Use the following procedure to prepare the diagram and upload evidence. Students who already know another diagramming tool may use it, but the exported evidence and source file must still be submitted.

1. Open <https://app.diagrams.net/> or the desktop version of draw.io / diagrams.net.
2. Choose Device, Google Drive, or OneDrive as the save location according to availability.
3. Create a blank diagram and name the file as CSA15_CO1_AT2_RegisterNumber_Concept_Map.drawio.
4. Place the capstone title at the center or as a clearly visible anchor node.
5. Add major cloud nodes: cloud definition, cloud evolution, cloud architecture, service models, deployment models, virtualization, web services, SOAP, REST, APIs, elasticity, scalability, resource pooling, measured service, and emerging paradigms.
6. Connect nodes with labeled arrows such as uses, depends on, enables, scales through, communicates through, belongs to, supports, provides, consumes, or controls.
7. Use colors meaningfully: one color for service models, one for cloud characteristics, one for API/web-service concepts, and one for capstone-specific elements.
8. Export the final diagram as PNG or PDF. Keep the editable .drawio source file also.
9. Paste the exported diagram image or screenshot in this DOCX under Task 1.
10. Upload the DOCX, exported PNG/PDF, and .drawio file to GitHub, then submit the GitHub link in Google Classroom.

GitHub Submission Procedure

Step	Required Action	Expected Evidence
1	Create or use your individual GitHub account.	GitHub profile link.
2	Create a repository named CSA15-CO-Assessments-RegisterNumber or use the repository instructed by faculty.	Repository URL.
3	Create a folder named CO1_AT2 inside the repository.	Folder visible in repository.
4	Upload this completed DOCX, exported diagram PNG/PDF, and editable .drawio source file.	At least three evidence files uploaded.
5	Add a short README.md explaining the capstone title and files submitted.	README visible in the AT2 folder.
6	Submit the GitHub folder link in Google Classroom.	GCR submission with repository/folder link.

Assessment Questions and Student Response Sections

Important

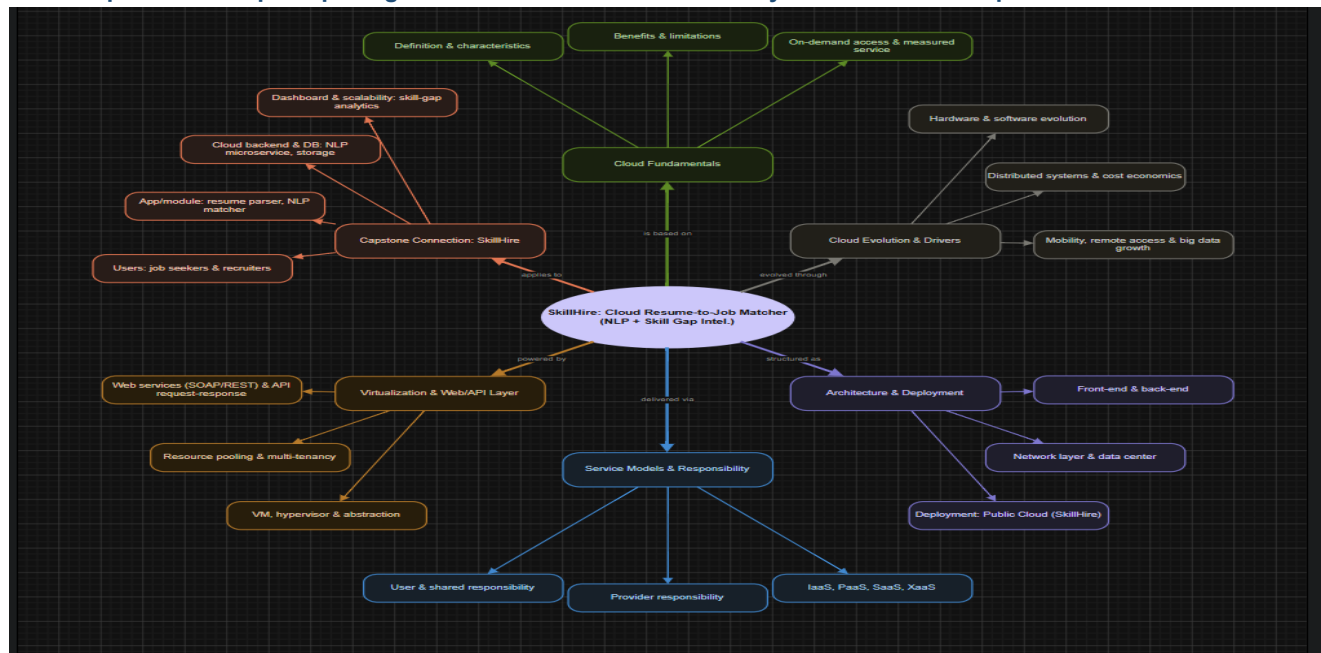
The capstone project may be developed by a team, but every response below must be written from the individual student's own perspective. Do not submit a common team answer.

Task 1 - Digital Cloud Concept Map

Create one integrated digital concept map for CO1. The map must show key nodes, link labels, relationship meaning, and a logical hierarchy. It must also connect the concepts to your selected capstone title from your individual perspective.

Mandatory Concept Group	Minimum Concepts / Nodes to Include
Cloud Fundamentals	Definition, characteristics, benefits, limitations, on-demand access, measured service.
Cloud Evolution and Drivers	Hardware evolution, internet software evolution, distributed systems, cost economics, mobility, remote access, big data growth.
Architecture and Deployment	Front-end, back-end, network layer, data center, management software, public/private/hybrid/community cloud.
Service Models and Responsibility	IaaS, PaaS, SaaS, XaaS, provider responsibility, user responsibility, shared responsibility.
Virtualization and Web/API Layer	Virtual machine, hypervisor, abstraction, resource pooling, multi-tenancy, web services, SOAP, REST, API request/response.
Capstone Connection	Your capstone users, app/module, cloud backend, database/storage, dashboard/analytics, scalability and elasticity idea.

Paste exported concept map image / screenshot here. Resize neatly within this white space.



The capstone anchor (SkillHire) sits at the top. Five CO1 concept pillars radiate from it, each linked with a meaningful verb.

Cloud Fundamentals (blue) — elasticity and on-demand access explain why SkillHire spins up NLP containers only on request, using measured service billing. Hybrid deployment keeps compute on AWS public cloud while sensitive resumes can use a private S3 bucket. Service Models (green) — IaaS (EC2) for the dev environment; PaaS (Lambda + API Gateway) for the NLP inference engine; SaaS-layer recruiter dashboards; XaaS extends to NLP-as-a-Service (AWS Comprehend / spaCy on Lambda). Shared responsibility: AWS owns runtime and hardware; team owns function code, IAM, and data governance. API / Web Layer (amber) — two primary REST endpoints: POST /upload (resume ingestion) and POST /match (skill-gap scoring). API Gateway enforces rate limits, OAuth validation, and CORS. SOAP is noted for legacy enterprise HR integrations.

Virtualization (purple) — Lambda containers isolate each NLP invocation. Docker images package spaCy, ensuring reproducibility and multi-tenancy without cross-tenant data leakage. Emerging Paradigms (red) — the NLP pipeline represents ML-as-a-Service. Future roadmap: edge inference for offline resume screening; Kinesis streaming for real-time job-board ingestion.

Link labels: enables, depends on, scales through, consumes, controls, provides.

Task 2 - Service Model Matrix

Complete the matrices by making technical decisions for your capstone project. Select only the rows that are relevant to your capstone design. If a row is not used, write a short technical reason instead of leaving it blank.

Your answer should show how responsibility shifts between cloud provider and student development team, and how the chosen services support the capstone workflow, API design, storage, security, scalability, and evidence submission.

Task 2A - Service Model Responsibility Matrix

Service Model	Capstone Role / Module	Provider Responsibility	Student Responsibility	Evidence to Submit
IaaS	Dev env, load-test VM, VPC	Hardware & hypervisor Physical DC security Network fabric	OS patching on EC2 Security Groups VPC/subnet config	EC2 screenshot, VPC diagram, Security Group export
PaaS	Lambda NLP functions, API Gateway	Managed runtime Lambda auto-patching Concurrency scaling TLS termination	Function code (spaCy) Deployment ZIP API route defs IAM execution role	Lambda code + ZIP, API Gateway config, CloudWatch log
SaaS	HR recruiter dashboard, report viewer	Hosting & uptime SLA CDN delivery App-level patching	Dashboard config Cognito user pools Report layout	Deployed React URL, Cognito config, dashboard screenshot
XaaS / Specialized Service	NLP-as-a-Service (AWS Comprehend / spaCy endpoint)	Model hosting GPU/CPU auto-alloc API throughput scaling	Resume JSON formatting Skill taxonomy mapping Post-processing logic	Comprehend API call log, skill-entity JSON, gap score module

Task 2B - Capstone Cloud Service Decision Matrix

Cloud Layer / Function	Selected Tool or Service	Data / Resource Handled	API or Integration Path	Security / Scaling Note
Client access and UI	React SPA – AWS Amplify + CloudFront	Resume upload, JD input, skill-gap dashboard	HTTPS GET/POST to API Gateway	CloudFront HTTPS; CDN auto-scales globally
API / backend logic	API Gateway + Lambda (Python, spaCy)	Resume text, JD text, skill lists, match score	POST /api/upload → PDF parse; POST /api/match → returns gap JSON	IAM role limits Lambda to S3+DynamoDB; 500 concurrent; provisioned concurrency
Authentication and identity	AWS Cognito + OAuth 2.0 JWT	User identity, role, session tokens	Amplify Auth SDK; JWT validated by Cognito Authorizer on API GW	MFA for recruiters; 1-hr token expiry; refresh rotation
Database / structured records	DynamoDB – on-demand mode	Match results, gap scores, user sessions	Lambda boto3 writes JSON; DynamoDB Streams trigger analytics	On-demand auto-scales; KMS encryption; PITR enabled
File / object storage	S3 private bucket (SSE-S3)	PDF resumes, gap reports, NLP model artefacts	Pre-signed URL upload (30-s expiry); signed URL download	Bucket blocks public access; versioning; 90-day lifecycle (GDPR)
Monitoring / logs / usage metrics	CloudWatch Logs + Alarms	Lambda logs, API latency, error rates	Lambda auto-publishes logs; custom NLP latency metric	Alarm → SNS if error rate >1% or P99 >3 s
Analytics / reporting output	QuickSight / recharts	Skill gap %, missing skills trend	QuickSight via Athena/S3; GET /api/report/{userId}	Row-level security per recruiter; auto-scales

Technical justification: Explain why the selected cloud service pattern fits your capstone project's users, data, API flow, scalability, and security needs.

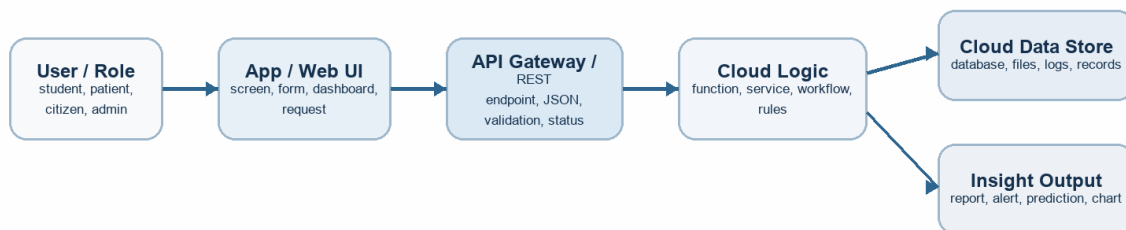
SkillHire's workload is bursty (many uploads during placement season, idle otherwise). Lambda (PaaS) scales from zero to hundreds of parallel NLP invocations in seconds and bills per millisecond, keeping idle cost near-zero. DynamoDB on-demand mirrors the same pattern for writes. Cognito offloads authentication security to the provider. S3 lifecycle rules and SSE encryption address GDPR for resume data. Together, this pattern removes all IaaS-level operational overhead so the team focuses entirely on NLP skill-gap logic.

Task 3 - Capstone Cloud Flow Mapping

Study the sample architecture-flow template below. Redraw it in draw.io / diagrams.net using your own capstone architecture. Replace every generic label with your actual user role, screen, API endpoint, backend logic, cloud storage, analytics output, security control, and scalability or elasticity decision.

Sample Capstone Cloud Flow Template

Replicate this structure using your own capstone screens, APIs, cloud services, data stores, and outputs.



Cross-cutting cloud decisions to label in your own diagram

Authentication / IAM

Who is allowed to access each API and data item?

Scalability

Which component must handle growing users or records?

Elasticity

Which component can expand or shrink based on demand?

Monitoring

What logs, usage metrics, or errors will be observed?

Security

Where are validation, privacy, backup, and access control applied?

Student instruction: redraw this flow in draw.io/diagrams.net and replace every label with your own capstone architecture.

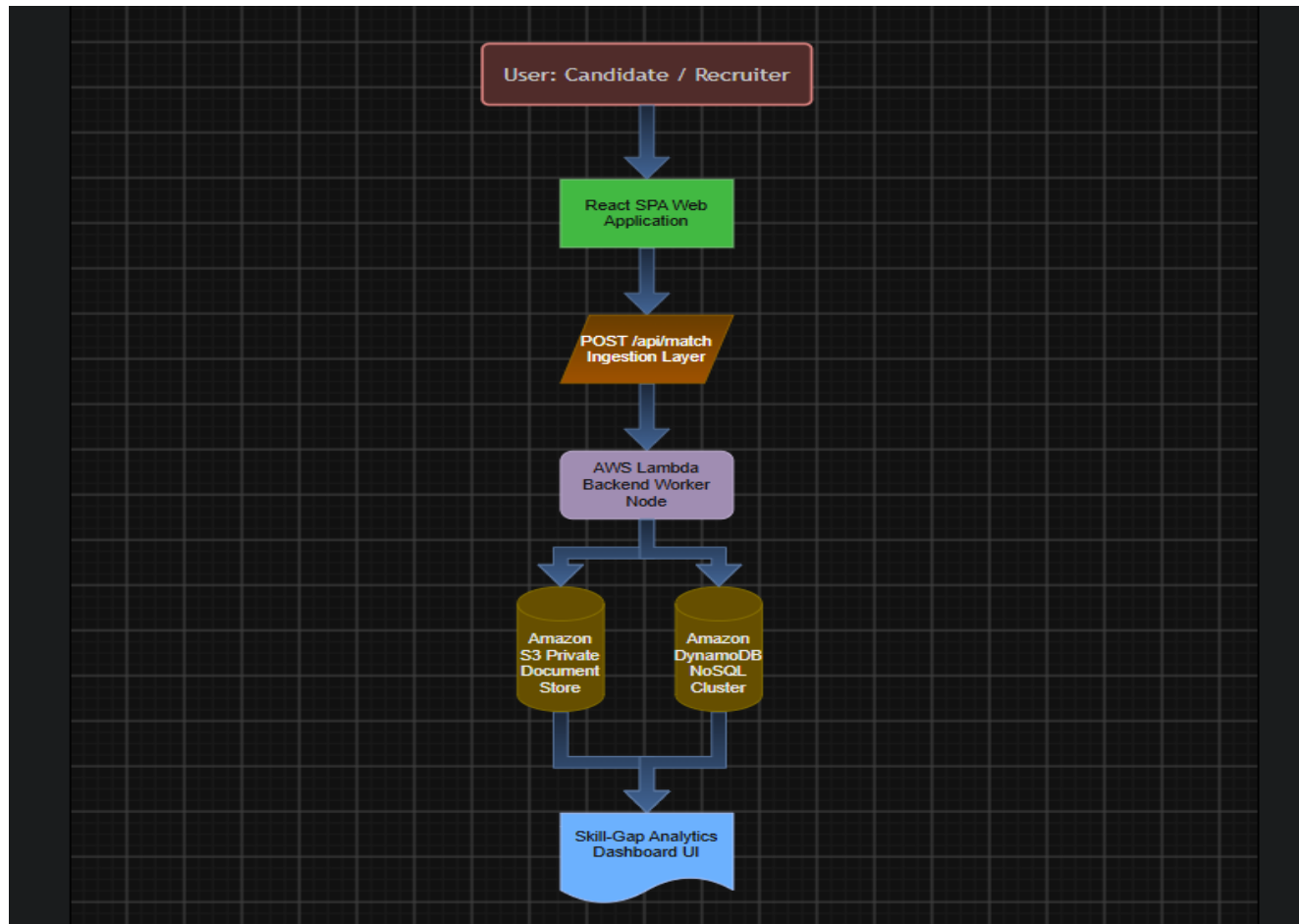
Student Drawing Requirement

Do not copy the sample as it is. Use it as a layout model, then create your own capstone-specific flow diagram with meaningful labels, arrows, data movement, API names, and cloud service choices.

Flow Component	Student Entry
User role and action	Candidate: log in → upload PDF resume. Recruiter: log in → paste job description. Both authenticate via Cognito OAuth 2.0 before any API call.
Frontend / app screen	React SPA (Amplify): Login → Cognito hosted UI Resume Upload → drag-and-drop PDF JD Input → text form Skill-Gap Dashboard → chart + skill list.
API endpoint and method	POST /api/upload → ingests PDF, converts to text, stores in S3, returns upload_id POST /api/match → {upload_id, job_description} → runs NLP, returns skill_gap JSON GET /api/report/{userId} → retrieves historical match results from DynamoDB
Backend processing rule	1. PDF→text via pdfplumber (Lambda) 2. spaCy NER extracts SKILL entities from resume + JD 3. skill_gap = JD_skills - Resume_skills; match_score = intersection / JD_skills × 100 4. Write result JSON to DynamoDB; store raw files in S3
Cloud DB / storage	S3 (private): /resumes/{userId}/{ts}.pdf; /reports/ DynamoDB SkillHireMatches: PK=user_id, SK=match_timestamp; fields: skill_gap_score, missing_skills[], matched_skills[]
Response / dashboard	{match_score:72, missing_skills:["Docker","K8s","CI/CD"], matched_skills:["Python","REST","SQL"]} Dashboard: bar chart of gap, missing skills with learning links

Security / Scalability / Elasticity	CloudWatch alarm → SNS if Lambda error rate exceeds threshold SECURITY: Cognito JWT on every call; S3 bucket policy blocks public; KMS at rest; VPC for Lambda↔DynamoDB SCALABILITY: API GW handles 10k+ RPS; Lambda scales to 500 concurrent; DynamoDB on-demand auto-adjusts ELASTICITY: Lambda containers spin up per request, terminate after — zero idle compute cost
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Paste your redrawn capstone cloud-flow diagram here.



Brief explanation: explain one complete request-response path from user action to cloud response.

A candidate logs into SkillHire via Cognito and receives a JWT. On the Upload screen, they submit their PDF; the React app sends POST /api/upload with the JWT in the Authorization header. API Gateway validates the token via the Cognito Authorizer and routes the request to the upload Lambda, which converts the PDF to text using pdfplumber, stores the file in the private S3 bucket, and returns an upload_id. The candidate then enters a job description and clicks Match; POST /api/match triggers the NLP Lambda, which runs spaCy NER on both the resume text and JD, computes the skill gap set and match score, writes the result to DynamoDB, and returns the JSON to the client. The React dashboard renders a bar chart of matched vs. missing skills. The full round-trip completes in under 2 seconds for a typical resume.

Task 4 - Individual Reflection

Answer the following from your own perspective. Avoid team-common wording.

1. What is the strongest cloud concept represented in your capstone project?

PaaS-based serverless computing (Lambda + API Gateway). The NLP engine must parse PDFs, run NER, and compute set operations — but only when a request arrives. Lambda encodes on-demand self-service, elasticity, and measured service directly into the architecture: containers instantiate on demand, run the inference, and release immediately. Without PaaS, the team would manage EC2, load balancers, and OS patches — all effort diverted from the NLP skill-gap logic that makes SkillHire unique.

2. Where will elasticity or scalability become important in your capstone product?

Two points: (1) campus placement season, when hundreds of students upload resumes simultaneously — Lambda absorbs the burst to 500 concurrent executions in seconds without pre-provisioning; (2) growth from a 50-student pilot to a cross-institution deployment — DynamoDB on-demand silently increases write throughput without any schema change. Elasticity also means cost efficiency in the off-peak trough, since both Lambda and DynamoDB charge per actual use, not reserved capacity.

3. Which API or web-service idea is most relevant to your project and why?

REST API with stateless request design. Each call to POST /api/match carries the JWT (identity), upload_id (resume reference), and job description — the server holds no session state, so any Lambda instance can serve any request, enabling effortless horizontal scaling. The API Gateway additionally acts as a policy enforcement point for rate limiting, CORS, and token validation, separating security concerns cleanly from the business logic inside Lambda.

4. What did you personally understand better after preparing this concept map and matrix?

Before this assessment, IaaS / PaaS / SaaS were abstract labels. After completing the Responsibility Matrix, I understood them as a spectrum of responsibility transfer: choosing Lambda (PaaS) meant explicitly delegating OS management, runtime patching, and capacity planning to AWS and accepting responsibility only for function code and IAM roles. The matrix made this trade-off concrete. The concept map also clarified the difference between cloud characteristics (elasticity — what cloud does) and mechanisms (Lambda auto-scaling — how SkillHire uses it), which I had previously conflated.

Evaluation Rubric - 100 Marks

Criteria	Marks	Expected Standard
Concept identification	20	Major CO1 concepts are accurately identified: cloud fundamentals, characteristics, evolution, service models, deployment models, virtualization, web services, SOAP/REST, APIs, elasticity, and scalability.
Relationship mapping	20	Links between concepts are accurate, meaningful, and labeled. The map should explain how concepts depend on, enable, consume, provide, scale, or control one another.
Hierarchy and organization	15	Concepts are arranged in a logical hierarchy with clear grouping, readable flow, and no disconnected keyword clusters.
Technical relevance	15	The map strongly reflects cloud course content and avoids decorative or generic non-technical content.
Service model matrix	15	IaaS, PaaS, SaaS, and XaaS are compared with provider responsibility, user responsibility, examples, capstone usage, and risks or limitations.
Capstone alignment and individual reflection	10	The student's own capstone interpretation is visible in the concept map, flow mapping, justification, and reflection. Team-common answers are avoided.
Visual clarity and digital evidence	5	Diagram is neat, readable, exported properly, and supported with GitHub evidence including source and exported files.

Faculty Evaluation Record

Evaluator Name		Marks Awarded	
Observation		Signature	